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When Second Order Shock Expansion Theory Breaks Down 04 June 2013

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Introduction

Experience with running the RS&T second order shock expansion theory code, SUPERSONIC BARROWMAN EQUATIONS.xls./xls reveal that occasionally it crashes, usually at a compression corner. The compression corner failures are due to a detached 2D shock at the corner. What's to be done? Under these circumstances, the user will accept an answer of lower quality rather than no answer at all. So we will revert to simpler theories, slender body or Newtonian to develop answers when the gold standard fails.

Analysis

The maximum stream deflection angle δ_{MAX} for locally sonic flow that an attached 2D shock can support for an upstream Mach number M_1 is given by eq. (169) of ref. (1):

$$\delta_{MAX} = \frac{1}{\sqrt{2}(\gamma+1)} \frac{(M_1^2 - 1)^{3/2}}{M_1^2} = 0.2946 \frac{(M_1^2 - 1)^{3/2}}{M_1^2} \qquad (169)$$

where γ = the ratio of specific heats for air = 1.4

Local supersonic flow is a necessary condition for the second order shock work expansion theory to work. Equation (169) was derived for Mach numbers not much greater than unity.

It is possible for the requirement for downstream supersonic flow to not be satisfied for larger Mach numbers and blunter shapes. When this happens the choice will be to either use slender body theory or Newtonian theory. The choice depends on the hypersonic similarity parameter. There are several alternative formulations of this, but the numerical results of ref. (2) show that best results can be obtained by using

$$H = \sqrt{M_1^2 - 1} \tan\delta, \qquad (1)$$

where H =the hypersonic similarity parameter

Then, in the event that the flow downstream of a compression corner is not supersonic, one evaluates H and follows the rule below:

If $H < 1$ then use slender body theory

If $H > 1$ then use Newtonian theory

Once one of these assignments has been made for a compression corner the entire body downstream of the corner is analyzed by the same approximation. Both theories are local in nature; the air loads on an element depend on only the local element geometry (and Mach number).

Slender Body Theory

Slender body theory as invented by Munk was documented in ref. (3). Suppose the body is slender of arbitrary slender longitudinal profile and circular cross section. Break the body into a sequence of conical frustums, and consider one such conical element.

Then the normal force coefficient slope for this element is

$$\frac{\partial C_N}{\partial \alpha} = 2(S_2 - S_1)/S_{REF}, \tag{2}$$

where C_N = Normal force coefficient,
α = Angle of attack, rad.,
S_1 = Body cross section area at the forward frustum face, ft^2,
S_2 = Body cross section area at the aft frustum face, ft^2, and
S_{REF} = Aerodynamic reference area, ft^2
Since body is assumed to have a circular cross section, the cross section areas both take the form, $S = \pi R^2$

The element center of pressure is given by

$$x_{CP} = (x_2 S_2 - x_1 S_1 - Vol)/(S_2 - S_1) \tag{3a}$$

if $S_2 \neq S_1$ and

$$x_{CP} = (x_2 + x_1)/2 \tag{3b}$$

if $S_2 = S_1$

Here, x_{CP} = Element center of pressure aft of the body datum location,
x_1 = Frustum front face distance aft of the body datum,
x_2 = Frustum aft face distance aft of the body datum, and
Vol = Frustum volume.

Note that slender body predicts zero wave drag on an element.

Modified Newtonian Theory

The standard Newtonian results for element normal force coefficient slope and center of pressure can be found in ref. (4). The correction for stagnation pressure can be found in ref. (5). Now, consider a conical frustum. Reference (5) gives the normal force coefficient slope as

$$\frac{\partial C_N}{\partial \alpha} = C_{PT} \frac{(S_2 - S_1)}{S_{REF}} \cos^2 \delta \, . \tag{4}$$

The frustum center of pressure is given by eq's. (3) as before.

References

1. "Equations, Tables, and Charts for Compressible Flow", Ames Research Staff, NACA Report 1135, 1953
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5. "Modified Newtonian Aerodynamics (rev.3)", C. P. Hoult, RST memo, 2013